

FPGA Video Processing Core

Real-time HDMI video pipeline on Spartan-7 implementing edge detection and LUT-based color grading at 1080p60.

VHDL	Vivado	HDMI	Spartan-7
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VERSION	STATUS	DEVICE	FORMAT	TOOLCHAIN	UPDATED
0.3.0	In Progress	XC7S50-1	1080p60	Vivado 2025.1	Sep 2026

1. Overview

The FPGA Video Processing Core is a fully pipelined HDMI video path implemented in VHDL on a Spartan-7. It receives a 1080p60 stream, converts it to a working colour space, applies a Sobel edge-detection stage and a LUT-based colour grade, and re-transmits over HDMI. Processing is line-buffered and streaming; no frame is ever stored, so added latency is measured in scan lines rather than frames.

The design target is deterministic real-time behaviour: at 148.5 MHz pixel clock the pipeline accepts one pixel per clock with no back-pressure, so throughput is fixed by the video timing rather than by data content. Effect parameters are writable at runtime through a register map over SPI, with updates taking effect on the next vertical blank to avoid tearing.

2. Datapath

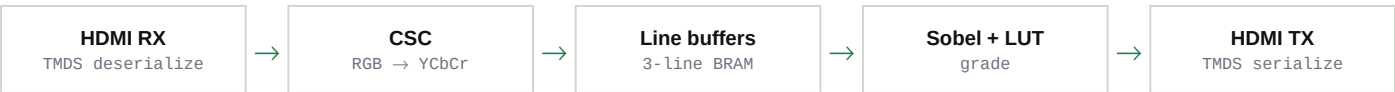


Figure 1 — Single clock domain after the RX deserializer. Total latency through the processing stages is 3 lines plus 11 pixel clocks.

The three-line buffer is what makes a 3×3 convolution possible without external memory. Two full lines plus the incoming line live in block RAM, presenting a sliding window to the Sobel stage each clock. Edge pixels at the frame boundary are handled by clamped replication rather than by a separate code path.

3. Module breakdown

hdmi_rx	TMDS deserialization, channel alignment, and timing recovery from the incoming stream.
csc	Fixed-point RGB to YCbCr conversion, BT.709 coefficients, two pipeline stages.
line_buffer	Dual-port BRAM ring holding three lines and exposing a 3×3 window per clock.
sobel	Separable 3×3 gradient with DSP-based magnitude approximation and threshold.
lut_grade	1D per-channel LUT in distributed RAM; runtime-loadable over the register interface.
hdmi_tx	TMDS encoding and 10:1 serialization with the OSERDESE2 primitives.

4. Implementation results

RESOURCE	USED	AVAILABLE	UTILISATION
Slice LUTs	8,412	32,600	25.8%
Slice registers	6,187	65,200	9.5%
Block RAM (36 Kb)	21	75	28.0%
DSP48E1	14	120	11.7%

Vivado 2025.1, XC7S50-1, default implementation strategy. Worst negative slack +0.42 ns at 148.5 MHz; the design meets timing with margin on the -1 speed grade.

5. Register map

```
-- SPI-accessible control registers, 16-bit address / 16-bit data
-- Writes latch at the next vertical blank unless IMMEDIATE is set.

0x0000 ID          r    = 0x5637 (device identity)
0x0004 CTRL        rw   [0] enable [1] bypass [2] immediate
0x0008 STATUS      r    [0] rx_lock [1] pll_lock [3:2] mode
0x0010 SOBEL_THRESH rw  unsigned 8.0, default 0x0028
0x0014 SOBEL_MIX    rw  unsigned 0.8, edge overlay blend factor
0x0020 LUT_ADDR     rw  entry index, 0..255
0x0024 LUT_DATA     rw  {R[7:0], G[7:0]} then B on second write
0x0030 FRAME_COUNT  r    increments on each vsync
0x0034 LINE_ERRORS  r    sticky; cleared by write
```

Listing 1 — Parameters update between frames, so a LUT reload never produces a torn frame.

6. Current status

- **Working.** RX lock, colour space conversion, line buffering, Sobel edge detection, and TX output are stable at 1080p60 on the reference board across extended runs.
- **In progress.** The colour grade is currently three independent 1D LUTs; the intended 3D LUT with tetrahedral interpolation is written but not yet closing timing at full rate.
- **Not started.** Audio passthrough, EDID emulation, and support for 1080p50 and 720p input modes, all of which are scoped but unimplemented.

7. Verification

- A VHDL testbench drives synthetic video timing and compares the Sobel output against a reference model, so gradient behaviour is checked per pixel rather than judged by eye.
- Hardware validation uses a pattern generator and a capture card, checking frame counts and line error counters over multi-hour runs to catch intermittent TMDS alignment loss.
- Timing closure is re-verified on every change to the DSP path; the 3D LUT work is currently blocked on exactly this gate.

8. Remaining work

MILESTONE	SCOPE	TARGET
3D LUT	Tetrahedral interpolation closing timing at 148.5 MHz.	Q4 2026
Mode support	1080p50 and 720p60 with automatic timing detection.	Q1 2027
1.0	Audio passthrough, EDID emulation, documented SPI driver.	Q2 2027